Deep Learning Based Object Detection in Low-Light and Blurred Images

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May 2026

Abstract

Low illumination and blur remain important challenges for deployed object detection systems. In this study, three pre-trained detectors—YOLOv8n, SSD MobileNet, and Faster R-CNN—are tested under sixteen controlled imaging conditions obtained from a 1000-image subset of the COCO 2017 validation set. Three intensity levels and three Gaussian blur kernel sizes are used to create synthetic degraded images. Three enhancement techniques—gamma correction, CLAHE, and combined Gamma+CLAHE—are applied to low-light images to assess recovery potential. Models are evaluated using COCO mAP, precision, recall, and F1-score without retraining. Under normal conditions, Faster R-CNN achieves the best detection accuracy (AP@50 = 0.577). YOLOv8n achieves the highest F1-score (0.526) and is the most robust model across degraded conditions. Gaussian blur causes the largest performance drop of all degradation types. CLAHE partially recovers performance under severe low-light; Faster R-CNN reaches AP@50 = 0.511 after enhancement. Results show that Faster R-CNN is most accurate on clean images, YOLOv8n is most robust on degraded images, and lightweight enhancement can restore detection without retraining.

Keywords: object detection, low-light imaging, Gaussian blur, YOLOv8, SSD MobileNet, Faster R-CNN, CLAHE, gamma correction, COCO, image degradation.

1 Introduction

Object detection is a computer vision task where the goal is not just to recognise what is in an image, but to find exactly where each object is by drawing a bounding box around it. In the last decade, deep learning has revolutionised this area. R-CNN, Fast R-CNN, and Faster R-CNN enabled much more accurate detection than previous work. Later, single-stage detectors such as SSD and YOLO made real-time detection on video streams possible.

1.1 Problem Statement

Nearly all these models are developed and evaluated on clean, well-lit images. This is not the case in the real world. Night-shot security cameras produce blurred and dark images. Rain causes dashcams to capture blurry pictures. Phones struggle in dark rooms. In such cases, models that perform well on clean images fail to detect or miss objects entirely.

This project addresses two of the most prevalent real-world degradations: low light and blur. Both are common in driving and surveillance settings, and both destroy the visual information that detection models rely on—edges, textures, and contrasts. The central question is: how much damage do these degradations cause, and can simple image enhancement repair it?

1.2 Proposed Approach

Three well-known pre-trained detectors are tested: **YOLOv8n**, the smallest and fastest version of YOLOv8, commonly used in real-time applications; **SSD MobileNet**, a lightweight single-stage detector designed for efficiency; and **Faster R-CNN**, a powerful two-stage detector that is slower but generally more accurate. All models use pre-trained COCO weights without retraining.

A set of 1000 images from the COCO 2017 validation dataset is used. Images are darkened at three levels (mild 0.70, medium 0.50, severe 0.30) and blurred with Gaussian kernels at three sizes (5×5, 11×11, 21×21), yielding sixteen image sets per model. Three enhancement methods—gamma correction, CLAHE, and Gamma+CLAHE—are applied to low-light sets. Performance is measured using AP@50, mAP, precision, recall, and F1-score.

1.3 Objectives

This project aims to: (1) measure exactly how much detection performance drops under low-light and blur degradation at different severity levels; (2) determine whether gamma correction, CLAHE, or a combination of both can restore lost performance without retraining; and (3) identify which detector architecture is most stable under challenging imaging conditions.

2 Related Work

2.1 Object Detection Methods

Traditional object detection relied on hand-designed features. Viola and Jones [2001] used a boosted cascade of Haar-like features for real-time face detection. Dalal and Triggs [2005] introduced Histograms of Oriented Gradients (HOG) as the standard feature for pedestrian detection.

Deep learning brought about a major change. R-CNN used region proposals and CNN-processed features to classify bounding boxes [Girshick et al., 2014]. Fast R-CNN ran the CNN only once on the entire image for efficiency [Girshick, 2015]. Faster R-CNN introduced a Region Proposal Network (RPN), yielding an end-to-end trainable model [Ren et al., 2015].

Single-stage detectors trade some accuracy for speed. YOLO casts detection as a regression problem [Redmon et al., 2016]; SSD uses multi-resolution feature maps to capture objects of different sizes [Liu et al., 2016]. YOLOv8, released in January 2023, uses a CSPDarknet backbone, feature pyramid neck, and an anchor-free head [Jocher et al., 2023]. A 2024 study on the ExDark dataset [Namana and Kumar, 2024] found YOLOv8m achieved mAP@0.50 = 0.886, outperforming Faster R-CNN (0.783) and SSD (0.693).

2.2 Low-Light and Blur Degradation

Low-light images reduce contrast, fade textures and edges, and add noise, causing small objects to disappear into the background. The ExDark dataset [Loh and Chan, 2019] provides over 7000 images across ten low-light scenarios for benchmarking detectors in dark conditions.

Blur from motion, camera shake, or defocus removes high-frequency features such as edges and textures. Deep deblurring networks such as DeblurGAN [Kupyn et al., 2018] and multi-scale CNNs [Nah et al., 2017] restore sharp images, but are slow and can introduce artefacts. This project instead evaluates detector robustness directly by blurring images with controlled Gaussian kernels.

2.3 Enhancement for Low-Light Detection

Gamma correction and CLAHE [Pizer et al., 1987] are fast, training-free techniques that increase contrast and reveal detail in dark regions. More complex deep learning methods—LLNet [Lore et al., 2017], RetinexNet [Wei et al., 2018], and Zero-DCE [Guo et al., 2020]—require large training datasets and increase model complexity. This project focuses on gamma correction and CLAHE to assess whether lightweight enhancement suffices.

2.4 Robustness Evaluation

Hendrycks and Dietterich [2019] benchmarked classifier robustness to corruptions and found significant accuracy drops. Michaelis et al. [2019] extended this to object detection, showing 10–20% lower mAP on common corruptions. These results motivate evaluating detectors on corrupted images beyond standard test sets.

2.5 Research Gap

Many papers address image enhancement without measuring its impact on detection, or evaluate a single model rather than comparing architectures. The ExDark 2024 study only compared YOLOv8 variants. There is a need to compare SSD MobileNet, Faster R-CNN, and YOLOv8n under identical low-light and blur conditions, and to assess whether simple enhancements suffice for recovery. This study addresses that gap.

3 Methodology

This project focuses on evaluation rather than training, built around six requirements: (R1) use a 1000-image COCO 2017 subset [Lin et al., 2014]; (R2) synthesise degraded images using low-light and Gaussian blur at three severity levels; (R3) enhance degraded images with gamma correction, CLAHE, and Gamma+CLAHE; (R4) detect using pre-trained models without retraining; (R5) compare all three detectors under identical conditions; and (R6) evaluate using mAP, AP@50, precision, recall, and F1-score.

Kanban was selected as the development methodology [Benson and Barry, 2011]. Waterfall was too rigid for an evolving research project; Scrum's team roles were unnecessary for a single researcher. Kanban provides a continuous-flow visual board with no fixed roles, allowing just-in-time planning. Tasks moved through "Ready", "Doing", and "Done" columns with work-in-progress limits.

Synthetic degradations enabled reproducible experiments at controlled severity levels. Deep enhancement networks and model training were excluded, as they require large datasets and GPU clusters and can introduce artefacts that complicate analysis.

4 Implementation

4.1 Dataset Preparation

All experiments used the COCO 2017 validation set. Processing all 5000 images across 16 conditions and three models would be prohibitively slow on a CPU. A subset of 1000 images was randomly selected using fixed random seed 42, ensuring reproducibility. Ground truth annotations were extracted and stored in `instances_val2017_1000.json` for use with `pycocotools`.

4.2 Image Degradation Pipeline

Both degradation types were applied at three severity levels, producing six degraded sets plus the original (seven sets per model).

4.2.1 Low-Light Simulation

Each pixel was multiplied by a brightness reduction factor: mild (0.70, 30% reduction), medium (0.50, 50% reduction), and severe (0.30, 70% reduction). Pixel values were clipped to $[0, 255]$ after multiplication.

4.2.2 Gaussian Blur Simulation

OpenCV's `GaussianBlur` was applied at three kernel sizes [Gonzalez and Woods, 2018]: mild (5×5, slight softening), medium (11×11, noticeable blur), and severe (21×21, strong blur with significant loss of detail). The standard deviation was set to zero, allowing OpenCV to derive it automatically from the kernel size.

4.3 Image Enhancement Pipeline

Three enhancement methods were applied to each of the three low-light sets, producing nine enhanced sets. Enhancement was not applied to blurred sets, as restoring blur requires deconvolution, which was outside scope.

4.3.1 Gamma Correction

Gamma correction applies a power-law transformation [Gonzalez and Woods, 2018]:

$$s = r^{\gamma} \tag{1}$$

where $r \in [0, 1]$ is normalised input intensity and $\gamma < 1$ brightens the image. A Look-Up Table (LUT) performs the per-pixel transformation efficiently:

$$\text{output} = \left(\frac{\text{input}}{255}\right)^{\gamma} \times 255 \tag{2}$$

Gamma values were matched to severity: mild ($\gamma = 0.8$), medium ($\gamma = 0.6$), severe ($\gamma = 0.4$).

4.3.2 CLAHE

CLAHE (Contrast Limited Adaptive Histogram Equalization) enhances local regions separately rather than globally [Pizer et al., 1987]. It was applied to the lightness channel (L) of the CIE L*a*b* colour space to avoid hue distortion. Parameters: clip limit = 2.0; tile grid size = 8×8.

4.3.3 Gamma + CLAHE Combined

Gamma correction was applied first to raise global brightness, followed by CLAHE to improve local contrast. This combined pipeline addresses both global and local characteristics of degraded images.

4.4 Model Implementation

All models used publicly available COCO pre-trained weights and a uniform confidence threshold of 0.25 across all 16 conditions (Table 1).

Table 1: Detection models used in this study.

Model	Library	Weights
YOLOv8n	Ultralytics	`yolov8n.pt`
SSD MobileNet	torchvision	SSDLite320_MobileNetV3
Faster R-CNN	torchvision	FasterRCNN_ResNet50_FPN

YOLOv8n [Jocher et al., 2023] was implemented via Ultralytics. The nano variant was chosen for CPU suitability; IoU threshold for NMS was set to 0.7.

SSD MobileNet [Liu et al., 2016, Howard et al., 2017] used `ssdlite320_mobilenet_v3_large` with DEFAULT pre-trained weights and the model's own preprocessing transform.

Faster R-CNN [Ren et al., 2015] used `fasterrcnn_resnet50_fpn` with DEFAULT pre-trained weights. Due to high compute requirements, it was run on Google Colab with a Tesla T4 GPU.

4.5 Evaluation Metrics

IoU measures bounding box overlap [Everingham et al., 2010]:

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}} \tag{3}$$

A prediction is a True Positive (TP) if IoU ≥ 0.5 with the correct class; otherwise it is a False Positive (FP). Unmatched ground truth boxes are False Negatives (FN) [Manning et al., 2008]:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

mAP was calculated via `pycocotools` following the official COCO protocol [Lin et al., 2014]:

$$\text{mAP} = \frac{1}{10} \sum_{\text{IoU} \in \{0.50, 0.55, \ldots, 0.95\}} \text{AP@IoU} \tag{6}$$

5 Results and Evaluation

5.1 Overall Performance

Table 2 summarises model performance across the four most critical conditions.

Table 2: AP@50 under key conditions for all three models.

Model	Normal	LL Sev.	Blur Sev.	CLAHE
YOLOv8n	0.443	0.407	0.226	0.414
SSD MobileNet	0.314	0.285	0.214	0.295
Faster R-CNN	0.577	0.506	0.245	0.511

Faster R-CNN achieved the highest AP@50 (0.577) and recall (0.593) under normal conditions, making it the most accurate model overall. YOLOv8n achieved the highest F1-score (0.526) and precision (0.693), making it the most balanced detector. SSD MobileNet ranked lowest across all metrics (AP@50 = 0.314 normal, average AP@50 = 0.277).

Three clear patterns emerged: (1) Gaussian blur caused the largest performance drop of all conditions; (2) low-light degradation was significantly less harmful than blur; and (3) CLAHE partially restored performance under severe low-light for YOLOv8n and Faster R-CNN, while SSD MobileNet showed minimal improvement from any enhancement.

5.2 Impact of Low-Light Degradation

Table 3: AP@50 across low-light severity levels.

Model	Norm.	Mild	Med.	Sev.	Drop
YOLOv8n	0.443	0.436	0.429	0.407	8.1%
SSD MobileNet	0.314	0.307	0.299	0.285	9.2%
Faster R-CNN	0.577	0.568	0.546	0.506	12.3%

Low-light degradation had a relatively moderate impact across all models (Table 3). Faster R-CNN showed the largest drop (12.3%), while YOLOv8n was the least affected (8.1%), suggesting YOLOv8n is the most robust model to brightness reduction.

5.3 Impact of Gaussian Blur

Table 4: AP@50 across Gaussian blur severity levels.

Model	Norm.	Mild	Med.	Sev.	Drop
YOLOv8n	0.443	0.415	0.344	0.226	48.9%
SSD MobileNet	0.314	0.298	0.270	0.214	31.8%
Faster R-CNN	0.577	0.509	0.394	0.245	57.5%

Gaussian blur produced the highest performance reduction of any condition (Table 4). Faster R-CNN dropped 57.5% (0.577 to 0.245) under severe blur; YOLOv8n dropped 49.0%. SSD MobileNet showed the lowest decrease (31.8%), though it also had the lowest baseline. Blur is a more fundamental issue than low light because it eliminates the sharp edges and textures that CNNs rely on to localise objects [Gonzalez and Woods, 2018].

5.4 Impact of Enhancement Methods

Table 5: AP@50 under enhancement methods (severe low-light).

Model	LL Sev.	Gamma	CLAHE	G+C	Norm.
YOLOv8n	0.407	0.382	**0.414**	0.401	0.443
SSD MobileNet	0.285	0.274	**0.295**	0.293	0.314
Faster R-CNN	0.506	0.472	**0.511**	0.507	0.577

CLAHE produced the strongest recovery across all models (Table 5). YOLOv8n improved from 0.407 to 0.414 with CLAHE, recovering 18.8% of the performance gap. Faster R-CNN improved from 0.506 to 0.511, recovering 7.0%. SSD MobileNet improved from 0.285 to 0.295, recovering 34.5% of the gap, though the absolute gain was small (0.010 AP@50).

Gamma correction alone *lowered* AP@50 for all three models, showing that global brightness adjustment is insufficient. The combined Gamma+CLAHE pipeline also failed to consistently outperform CLAHE alone; YOLOv8n reached 0.414 with CLAHE but only 0.401 with Gamma+CLAHE. A likely cause is that gamma correction excessively brightens certain areas before CLAHE is applied, eliminating useful local contrast. Overall, CLAHE was the most effective enhancement, though recovery was limited and normal-condition performance was not fully restored.

5.5 Requirements Evaluation

All must-have and should-have requirements were fulfilled. R1 (low-light robustness) and R2 (blur robustness) were fully met with measured AP@50 drops across all models. R3 (enhancement) was *partially met*: CLAHE provided measurable but incomplete recovery, and no method fully restored normal-condition performance. R4 (fair comparison) was met: all models used the same 1000-image subset, confidence threshold, and evaluation protocol. R5 (best model identification) was met: Faster R-CNN was best on AP@50; YOLOv8n was best on F1-score balance.

6 Conclusion

6.1 Summary of Findings

This study evaluated YOLOv8n, SSD MobileNet, and Faster R-CNN on images degraded by low light and blur using 1000 COCO 2017 images. Key findings are:

- Faster R-CNN is the most accurate model on clean images (AP@50 = 0.577), and partially recovers under CLAHE (AP@50 = 0.511).
- YOLOv8n is the most balanced and robust detector (F1 = 0.526, precision = 0.693) across degraded conditions.
- SSD MobileNet performed worst across all metrics despite having the highest mean confidence scores, indicating that confidence scores are not a reliable indicator of detection quality.
- Gaussian blur is more harmful than low-light degradation, as it destroys the edges and textures CNNs rely on for object localisation.
- CLAHE outperforms gamma correction and Gamma+CLAHE for all models. Local contrast enhancement is more effective than global brightness adjustment.

6.2 Recommendations

Three improvements could strengthen future work. First, the 1000-image random selection may have over-represented some categories; a balanced selection across all 80 COCO categories would improve reliability [Lin et al., 2014]. Second, learning-based enhancement methods such as Zero-DCE [Guo et al., 2020] or RetinexNet [Wei et al., 2018] may achieve better recovery under severe degradation. Third, per-category analysis would reveal which object types are most affected, providing more targeted insights for real-world applications.

6.3 Future Work

Future work should evaluate the same models on real photographs taken in truly dark or blurry conditions to validate whether the synthetic-degradation patterns generalise [Sakaridis et al., 2018]. Integrating enhancement and detection into a single end-to-end trainable pipeline would likely improve over the sequential approach used here. Testing larger YOLOv8 variants and architectures such as RT-DETR [Zhao et al., 2023] would determine whether more powerful models are inherently more robust. Adding noise, fog, rain, and compression artefacts would further increase evaluation realism.

References

Benson, J. and Barry, T. (2011) *Personal Kanban: Mapping Work, Navigating Life*. Modus Cooperandi Press.

Dalal, N. and Triggs, B. (2005) Histograms of oriented gradients for human detection. *CVPR*, pp. 886–893.

Everingham, M. et al. (2010) The Pascal VOC challenge. *Int. J. Computer Vision*, 88(2), pp. 303–338.

Girshick, R. et al. (2014) Rich feature hierarchies for accurate object detection. *CVPR*, pp. 580–587.

Girshick, R. (2015) Fast R-CNN. *ICCV*, pp. 1440–1448.

Gonzalez, R.C. and Woods, R.E. (2018) *Digital Image Processing*, 4th edn. Pearson.

Guo, C. et al. (2020) Zero-reference deep curve estimation for low-light image enhancement. *CVPR*, pp. 1780–1789.

Hendrycks, D. and Dietterich, T. (2019) Benchmarking neural network robustness to common corruptions. *ICLR*.

Howard, A.G. et al. (2017) MobileNets: Efficient CNNs for mobile vision. *arXiv*:1704.04861.

Jocher, G., Chaurasia, A. and Qiu, J. (2023) *Ultralytics YOLOv8*. `https://github.com/ultralytics/ultralytics`.

Kupyn, O. et al. (2018) DeblurGAN: Blind motion deblurring using conditional adversarial networks. *CVPR*, pp. 8183–8192.

Lin, T.Y. et al. (2014) Microsoft COCO: Common objects in context. *ECCV*, pp. 740–755.

Liu, W. et al. (2016) SSD: Single shot multibox detector. *ECCV*, pp. 21–37.

Loh, Y.P. and Chan, C.S. (2019) Getting to know low-light images with the ExDark dataset. *Computer Vision and Image Understanding*, 178, pp. 30–42.

Lore, K.G., Akintayo, A. and Sarkar, S. (2017) LLNet: A deep autoencoder approach to low-light enhancement. *Pattern Recognition*, 61, pp. 650–662.

Manning, C., Raghavan, P. and Schütze, H. (2008) *Introduction to Information Retrieval*. Cambridge University Press.

Michaelis, C. et al. (2019) Benchmarking robustness in object detection. *arXiv*:1907.07484.

Nah, S., Kim, T.H. and Lee, K.M. (2017) Deep multiscale CNN for dynamic scene deblurring. *CVPR*, pp. 3883–3891.

Namana, M.S.K. and Kumar, B.U. (2024) Night-time surveillance object detection with YOLOv8. *Int. J. Safety and Security Eng.*, 14(6), pp. 1763–1773.

Pizer, S.M. et al. (1987) Adaptive histogram equalization and its variations. *Computer Vision, Graphics, and Image Processing*, 39(3), pp. 355–368.

Redmon, J. et al. (2016) You only look once: Unified real-time object detection. *CVPR*, pp. 779–788.

Ren, S. et al. (2015) Faster R-CNN: Towards real-time object detection with region proposal networks. *NeurIPS*, 28, pp. 91–99.

Sakaridis, C., Dai, D. and Van Gool, L. (2018) Semantic foggy scene understanding with synthetic data. *Int. J. Computer Vision*, 126(9), pp. 973–992.

Viola, P. and Jones, M. (2001) Rapid object detection using a boosted cascade of simple features. *CVPR*, vol. 1, pp. 511–518.

Wei, C. et al. (2018) Deep Retinex decomposition for low-light enhancement. *BMVC*.

Zhao, L. et al. (2023) DETRs beat YOLOs on real-time object detection. *arXiv*:2309.05694.